(3) Implement a given strategy to determine a mathematical solution

Although the problem statement does not explicitly state "use trigonometry," the diagram (45° angle, vertical height 150 m) and the structure of the problem clearly direct the solver to use sine or cosine relationships in a right triangle. This aligns with "implementing a given strategy" within a defined geometric context.

(6) Develop mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

The problem includes a schematic diagram (not to scale). The test-taker must **extract geometric information** from it: a right triangle with known **opposite side (height) = 150 m**, **angle = 45°**, and must find the **hypotenuse (rope length)**. This directly matches "extracting mathematical information from diagrams."

(7) Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

This problem involves **translating geometric relations into algebra**:

$$\sin 45^\circ = \frac{150}{L} \quad \text{or} \quad \cos 45^\circ = \frac{150}{L}$$

and requires numerical computation:

$$L = \frac{150}{\sin 45^\circ} \approx \frac{150}{0.7071} \approx 212 \text{ m}$$

This constitutes working with geometric and algebraic representations.

(12) Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

The solution involves multiple steps:

1. Recognizing the right triangle model;
2. Selecting the appropriate trigonometric function (sin or cos);
3. Substituting known values;
4. Performing the calculation;
5. Rounding and matching the answer choice.

(13) Use an understanding of the context to guide or expedite the mathematical solving process

The context—"kite rope pulling the ship at a 45° angle"—helps the solver understand why a right triangle model is appropriate and prompts the use of **special trigonometric values for 45°** ($\sin 45^\circ = \cos 45^\circ \approx 0.7071$), which can expedite the calculation.

(1) Perform a simple calculation

The final computation $150/0.7071 \approx 212$ is a simple arithmetic operation, though it relies on prior understanding of trigonometric values.

(3) Implement a given strategy to determine a mathematical solution

题目虽未在题干明确给出“使用三角函数”的策略，但图示（45°角、垂直高度 150m）与问题结构明确指向利用直角三角形中的正弦或余弦关系求解，属于在给定几何情境下实施明确的数学策略。

(6) Develop mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

题目附有示意图（虽未按比例绘制），考生必须从图中**提取几何信息**：一个直角三角形，其中已知**对边（高）**=150 m，**角度**=45°，需要求**斜边（绳长）**。这完全符合“从数学图形中提取信息”这一维度。

(7) Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

本题涉及**几何关系的代数化**：

$$\sin 45^\circ = \frac{150}{L} \quad \text{或} \quad \cos 45^\circ = \frac{150}{L}$$

并要求进行数值运算：

$$L = \frac{150}{\sin 45^\circ} \approx \frac{150}{0.7071} \approx 212 \text{ m}$$

这属于对几何表示与代数表达式的综合运用。

(12) Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

解题是多步的：

1. 识别直角三角形模型；
2. 选择合适的三角函数（sin 或 cos）；
3. 代入已知值；
4. 进行计算；
5. 四舍五入并匹配选项。

(13) Use an understanding of the context to guide or expedite the mathematical solving process

情境是“风筝绳以 45° 角牵引船只”，这帮助考生理解为何使用直角三角形模型，并确定使用**45°角的特殊三角函数值**（$\sin 45^\circ = \cos 45^\circ \approx 0.7071$），从而加快计算。

(1) Perform a simple calculation

最终计算 $150/0.7071 \approx 212$ 是简单算术，但依赖于对三角函数的理解。